

Experiment-03

Study of VTC of Op-Amp Comparator, Non-Inverting Amplifier & Inverting Amplifier and Study of I-V Characteristics of Diode

CSE251 - Electronic Devices and Circuits Lab

Objective

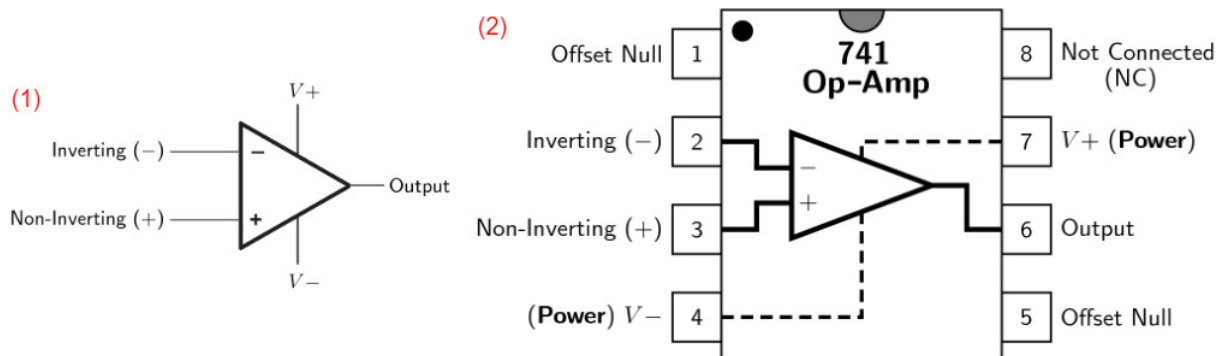
1. To understand the VTC of various Operational Amplifier (Op-Amp) based circuits
2. To become familiar with a silicon p-n junction diode and understand its operation
3. To study the current-voltage characteristics, i.e. I-V characteristics of silicon p-n junction diodes

Equipment

1. p-n junction diode (1N4007)
 2. Op-Amp (uA741)
 3. Resistance ($1k\Omega$, $2.7k\Omega$)
 4. Function Generator
 5. Oscilloscope
 6. Breadboard, Wires
 7. Digital Multimeter
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Background Theory

Op-Amp Circuit & IC Pin Diagram



(1) Op-Amp Simplified Circuit Symbol (2) Op-Amp IC Pin Diagram

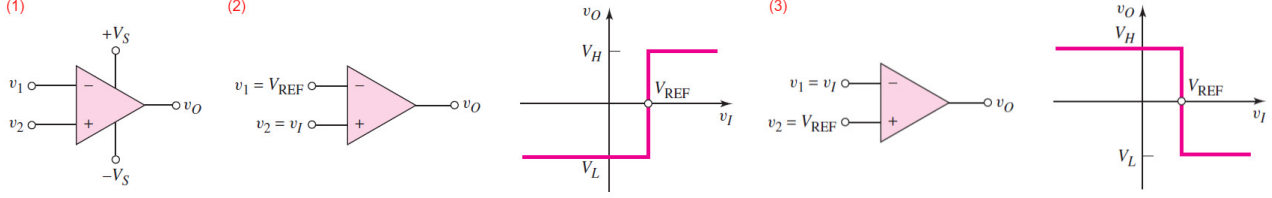
Voltage Transfer Characteristics (VTC)

Voltage Transfer Characteristics (VTC) refers to the relationship between the output voltage and the input voltage of a circuit. It is a crucial concept in circuit design. The VTC curve plots the output voltage against the input voltage, revealing how a circuit responds to varying input signals.

VTC of Op-Amp Comparator

The comparator is essentially an op-amp operated in an open-loop configuration, as shown below:

A comparator compares two voltages to determine which one is larger. Comparator is usually biased at voltages V_S^+ and V_S^- , although other biases are also possible. When, non-inverting input $>$ inverting input then, $v_O = V_S^+$. When, inverting input $>$ non-inverting input, i.e. $v_1 > v_2$ then, $v_O = V_S^-$. The figures below show two comparator configurations along with their voltage transfer characteristics (VTC) to illustrate the behaviour of a comparator with V_{REF} as reference voltage, which can be controlled to get the desired output.

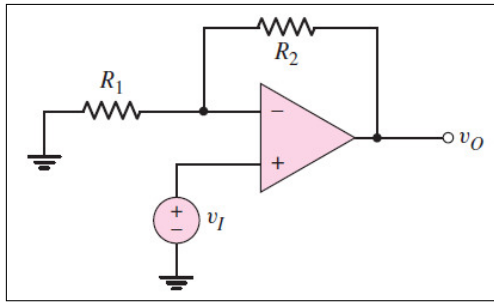


(1) Op-Amp Comparator (2) Non-inverting Circuit (3) Inverting Circuit

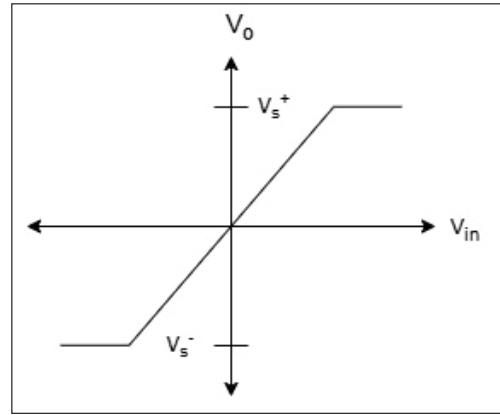
VTC of Non-Inverting Amplifier

The following equation shows the relationship between the input and output of a non-inverting amplifier, and the following diagrams show a non-inverting amplifier and its VTC:

$$v_O = \left(1 + \frac{R_2}{R_1}\right) \times v_I; \text{ where, gain} = \left(1 + \frac{R_2}{R_1}\right)$$



Non-inverting Amplifier

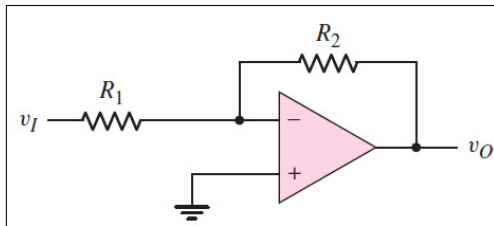


VTC of Non-inverting Amplifier

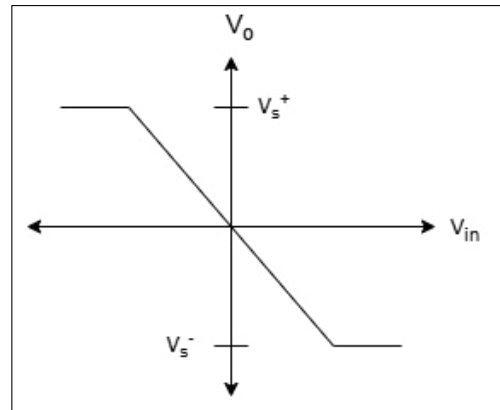
VTC of Inverting Amplifier

The following equation shows the relationship between the input and output of an inverting amplifier, and the following diagrams show an inverting amplifier and its VTC:

$$v_O = -\left(\frac{R_2}{R_1}\right) \times v_I; \text{ where, gain} = -\frac{R_2}{R_1}$$



Inverting Amplifier



VTC of Inverting Amplifier

Task-01: VTC of Op-Amp Comparator

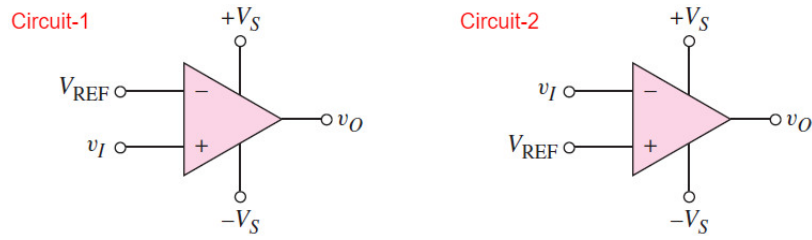
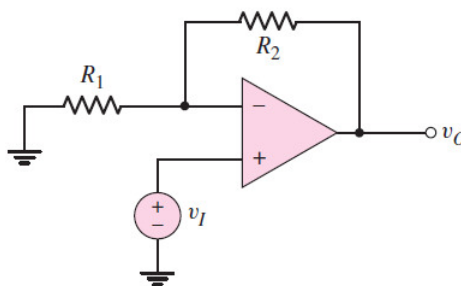


Figure 7

Procedure

1. Construct **Circuit-1** with $v_I = 10$ V (p-p), 1 kHz sine wave and $V_{REF} = 2$ V. Use the trainer board for the supply voltages, $V_S^+ = +8V$ and $V_S^- = -8V$.
2. The ground of the oscilloscope, trainer board and function generator should be connected.
3. Connect CH1 and CH2 of the oscilloscope to v_I and v_O respectively.
4. For Observing VTC we need to go to the XY mode:
 - (a) Press the **Autoset** button → Push the **Position** knobs of both channels (i.e. push to zero).
 - (b) Press the **Acquire** button → Press the **XY** button which can be found below the display → Press the **Triggered XY** button which can be found on the right side of the display.
 - (c) Change the scaling and position of the plot using the **Scale** knob and **Position** knob of both channels respectively if you need.
 - (d) Observe the VTC graph and capture it along with the measurements using a camera.
 - (e) Increase the V_{REF} and set it to 4 V, then observe the VTC graph and capture it with a camera.
 - (f) Next, decrease the V_{REF} and set it to 0 V, observe the VTC graph and capture it using a camera.
5. Now, construct **Circuit-2** and repeat the experiment with the same values given above and observe the VTC graph and capture it with a camera.

Task-02: VTC of Non-Inverting Amplifier

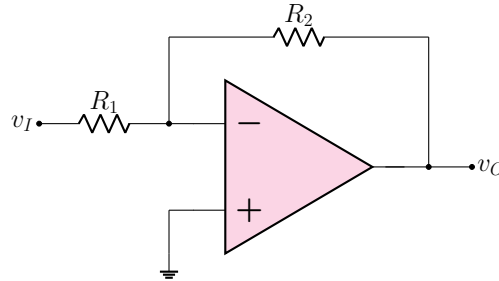


Procedure

1. Construct the circuit with $v_I = 10$ V (p-p), 1 kHz sine wave. Use $R_1 = 1$ k Ω , $R_2 = 2.7$ k Ω . Use the supply voltages, $V_S^+ = +8V$ and $V_S^- = -8V$.
2. The ground of the oscilloscope, trainer board, and function generator should be connected.
3. Connect the CH1 and CH2 of the Oscilloscope to v_I and v_O respectively. Use the **Scale** knob to ensure that the scales of CH1 and CH2 are the same.
4. For Observing VTC we need to go to the XY mode:
 - (a) Press the **Autoset** button → Push the **Position** knobs of both channels (i.e. push to zero).
 - (b) Press the **Acquire** button → Press the **XY** button which can be found below the display → Press the **Triggered XY** button which can be found on the right side of the display.

- (c) Change the scaling and position of the plot using the **Scale** knob and **Position** knob of both channels, respectively if you need.
- (d) Observe the VTC graph and capture it with a camera.

Task-03: VTC of Inverting Amplifier



Procedure

1. Construct the circuit with $v_I = 10\text{ V}$ (p-p), 1 kHz sine wave. Use $R_1 = 1\text{ k}\Omega$, $R_2 = 2.7\text{ k}\Omega$. Use the supply voltages, $V_S^+ = +8\text{ V}$ and $V_S^- = -8\text{ V}$.
2. The ground of the oscilloscope, trainer board, and function generator should be connected.
3. Connect the CH1 and CH2 of the Oscilloscope to v_I and v_O respectively. Use the **Scale** knob to ensure that the scales of CH1 and CH2 are the same.
4. For Observing VTC we need to go to the XY mode:
 - (a) Press the **Autoset** button → Push the **Position** knobs of both channels (i.e. push to zero).
 - (b) Press the **Acquire** button → Press the **XY** button which can be found below the display → Press the **Triggered XY** button which can be found on the right side of the display.
 - (c) Change the scaling and position of the plot using the **Scale** knob and **Position** knob of both channels respectively if you need.
 - (d) Observe the VTC graph and capture it with a camera.

Background Theory

I-V Characteristics

I-V characteristic defines the relationship between current flow and voltage across two terminals of a device or element. It is a tool for understanding the operation of the circuit. The I-V characteristics are found by evaluating the response of a device/element under different conditions. The behavior of a device depends on the applied excitation and can change if the excitation changes. For example, a device may act as an “open circuit” under certain input conditions and as “voltage source” in another. A diode acts as an open circuit below a specific threshold voltage and acts differently beyond it.

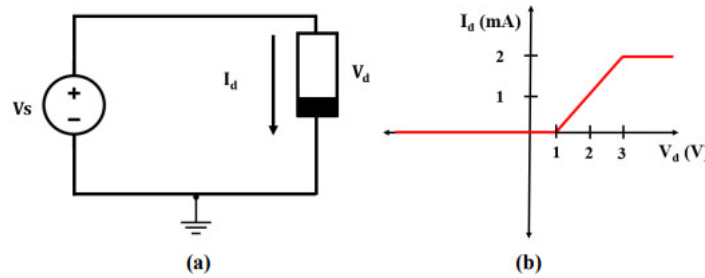


Figure 8: (a) A circuit with a voltage source, (b) I-V characteristic of the device

A simple circuit with a voltage source and an electronic device is shown in the figure above. The voltage source acts as an excitation medium for the device. Varying the voltage source would result in change in the current flow, I_d across the device. By plotting this current with respect to the voltage across the device, V_d , the I-V characteristics of this device can be determined.

I-V characteristics illustrates the behaviour of a device which is enough to know the device. Depending on the I-V characteristics, the devices can be divided into two categories:

- (1) Linear Devices
- (2) Non-linear Devices

If the current through a device is a linear function of the applied voltage across it, it is a linear device.

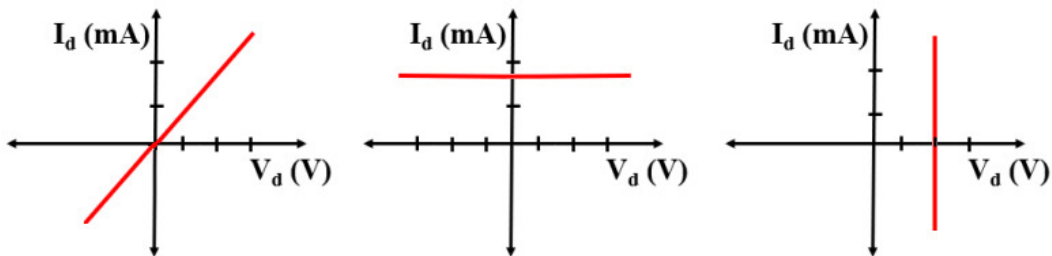


Figure 9: I-V characteristics of some linear elements

If the current through a device is a non-linear function of the applied voltage across it, it is a non-linear device.

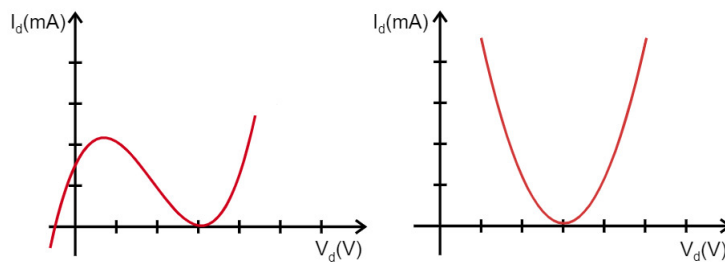


Figure 10: I-V curves of some non-linear elements

In this experiment, we will study the I-V characteristics of diodes which is an electronic device. We will observe that, diodes exhibit non-linear I-V characteristics.

Introduction to Diode

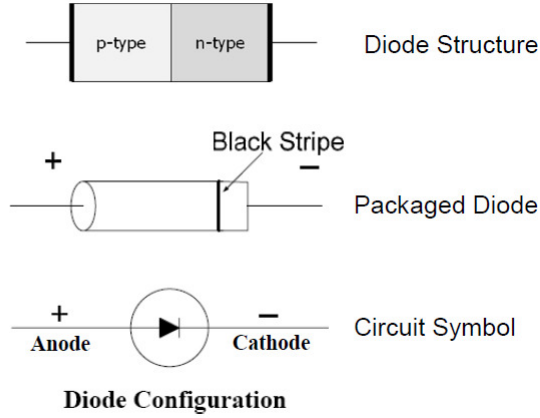


Figure 11: *Diode Structure, Packaged Diode and Circuit Symbol*

Diode is a semiconductor device that allows current flow only in one direction, from p to n or anode to cathode. The schematic diagram, diode notation and circuit symbol are shown in the figure above. Diodes are usually marked with a dot or a bar appearing on the cathode side. This mark helps identify the diode terminals.

I-V Characteristics of a Real Diode

Real diodes are made of semiconductor materials, which have highly non-linear I-V characteristics. However, under certain conditions and approximations, the real diode behaves like an ideal diode.

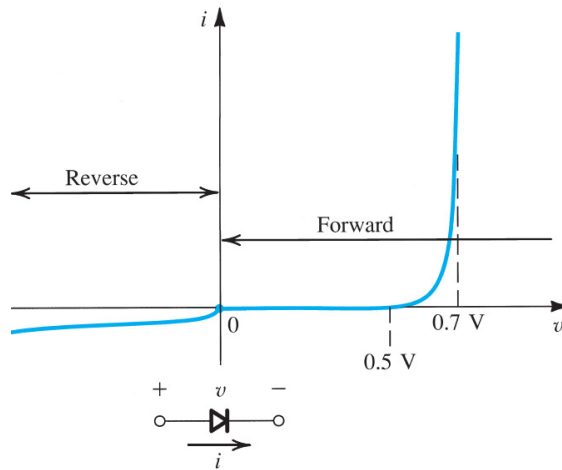


Figure 12: *I-V Characteristics of a Real Diode*

For a real diode, the current I_D when a voltage V_D is applied across it is given by:

$$I_D = I_S \left(\exp \left(\frac{V_D}{nV_T} \right) - 1 \right)$$

Here,

- I_S is called the reverse saturation current
- n is called the diode ideality factor, and it has a value between 1 and 2
- V_T is called the thermal voltage, which has a value of 25 mV at 27° C

However, there are 2 special cases:

- When $V_D \gg nV_T$: in this case, $\exp(V_D/nV_T)$ will be much higher than 1, and hence we can ignore 1. The equation becomes, $I_D \approx I_S \exp(V_D/nV_T)$
- When $V_D < 0$: in this case, $\exp(V_D/nV_T)$ will be negligible compared to 1, and hence we can ignore the $\exp(V_D/nV_T)$ term and the equation becomes, $I_D \approx -I_S$

Diode Resistance

As the diode I-V characteristic is not linear, it will have different resistances at different points on the curve. A dynamic or AC resistance for the diode is defined as,

$$r_d = \frac{d}{di}(v) \approx \frac{nV_T}{I_D}$$

The static or DC resistance at any point is defined as, $R_D = V_D/I_D$

Diode Specification

There are many specifications for each type of diode, the most important two are:

1. Peak inverse Voltage (PIV): maximum voltages the diode can tolerate in reverse direction.
2. Maximum Forward Current (I_F) the maximum current the diode can conduct in forward biased condition without exceeding the safe limit.

Diodes are widely used in applications such as mixers, detectors, protection circuits. In this experiment you will investigate its I-V characteristics.

I-V Characteristics of an Ideal Diode

Ideally, we want a diode to behave like an electronic valve. It allows any amount of current in one direction, while blocking all the currents in the opposite direction. This behavior can be characterized using the current and voltage relation between the diode (or for any electronic device) which is also known as the I-V characteristics of the device.

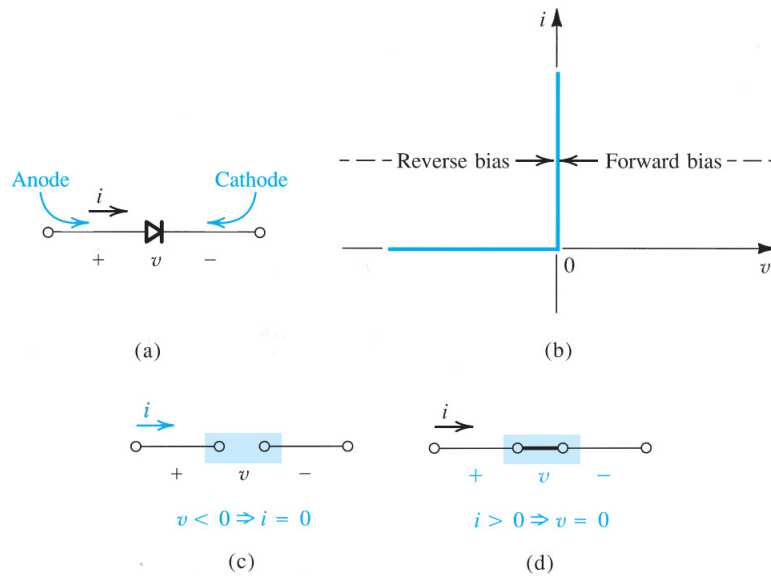
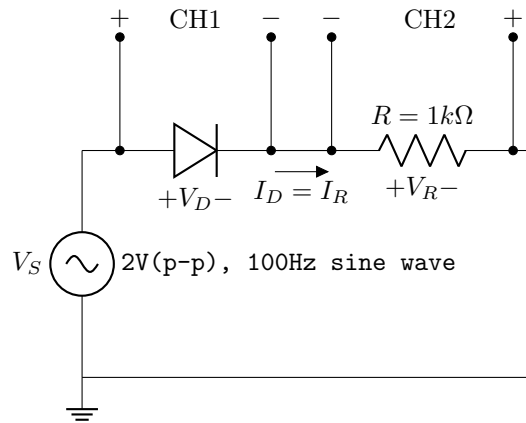


Figure 13: *I-V Characteristics of an ideal diode*

Task-04: Diode I-V Characteristics



Procedure

1. Construct the circuit given above. Use the function generator to generate a 2V(p-p), 100Hz sine wave for the supply voltage V_S .
2. Connect the CH1 and CH2 of the oscilloscope to the circuit as shown in the figure above.
3. Observe the I-V characteristics of the diode in the XY mode of the oscilloscope and capture the image. To use the XY mode:
 - (a) Press the **Autoset** button → Push the **Position** knobs of both channels (i.e. push to zero).
 - (b) We need to invert the CH2 in the oscilloscope due to its inverted connection to the circuit. To do so: Press **CH2** button → Press the **Invert** button which can be found on the bottom of the display of the oscilloscope to select the option **On**.
 - (c) Press the **Acquire** button → Press the **XY** button which can be found below the display → Press the **Triggered XY** button which can be found on the right side of the display.
 - (d) Change the scaling and position of the plot using the **Scale** knob and **Position** knob of both channels, respectively if you need.

You will see a small screen showing the I-V characteristics graph using the XY mode of the oscilloscope. The XY mode plots the voltage data of CH1 and CH2 in the x-axis and y-axis respectively. So, the x-axis represents V_D . As, $I_D = I_R \propto V_R$, the y-axis represents I_D .

- (e) Observe the I-V graph and capture it with a camera.

Signature of the lab faculty

Report Submission Guidelines

1. Add a cover page that includes course code, course title, name, student ID, group, semester, date of performance, date of submission.
2. Attach the captured photos of all the waveforms and measurements you have observed in the Oscilloscope. Each photo should contain necessary description.
3. Answer the questions in the “Test Your Understanding” section.
4. Add a brief Discussion regarding the experiment. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
 - What did you learn from this experiment?
 - What challenges did you face and how did you overcome the challenges? (if any)
 - What mistakes did you make and how did you correct the mistakes? (if any)
 - How will this experiment help you in future experiments of this course?

Test Your Understanding

Answer the following questions:

1. $R = 1k\Omega$ was used in the experiment. If we use $R = 2.2k\Omega$, will there be any problem in observing the I-V characteristics plot? Explain briefly.

Answer:

2. From the I-V characteristics of a diode that you obtained, which devices can be used to model the diode?

Answer:

3. Discuss the differences in behavior between the comparator's switching operation and the amplifier's linear scaling. For the comparator, use a 4 V (peak-to-peak) sine wave to the inverting terminal with a fixed reference of -1 V to the non inverting terminal. For the non-inverting amplifier, set the gain to 2 with a 1 V (peak-to-peak) sine wave input. For each circuit, draw the VTC (i.e., v_{out} vs. v_{in}) on separate graphs with proper labels and discuss accordingly.

Answer: